

$$T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = 3a - 4b + c - d$$

Escrever 2 matrices quaisquer

$$M = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad N = \begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}$$

Testar aditividade:

$$T(\underline{M+N}) = T(\underline{M}) + T(\underline{N})$$

$$\begin{aligned} T(M+N) &= T\left(\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}\right) \\ &= T\left(\begin{bmatrix} 7 & 3 \\ 1 & 1 \end{bmatrix}\right) = 3 \times 7 - 4 \times 3 + 1 - 1 = \\ &= 21 - 12 + 1 - 1 = \\ &= 9 \end{aligned}$$

$$\begin{aligned} T(M) + T(N) &= T\left(\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}\right) + T\left(\begin{bmatrix} 5 & 2 \\ 1 & 0 \end{bmatrix}\right) \\ &= (3 \times 2 - 4 \times 1 + 0 - 1) + (3 \times 5 - 4 \times 2 + 1 - 0) \\ &= (6 - 4 - 1) + (15 - 8 + 1) \\ &= 1 + 8 = 9 \\ &= 9 \checkmark \end{aligned}$$

linear